

A.1.2 Solutions

1. The main idea is to use the function `cumsum()` (cumulative sum) to compute the cumulative counts of head. Dividing (componentwise) by the sequence `1:n` gives the relative frequencies for each k .

```
n <- 10000 # given
X <- sample(x = c(0, 1), size = n, replace = TRUE) # given

# compute relative frequencies
relative_frequency <- cumsum(X)/(1:n)

# plot relative frequencies
plot(relative_frequency, type = "l", ylim = c(0, 1),
      xlab = "number of trials",
      ylab = "relative frequency")
abline(h = 0.5, lty = 2, col = "grey") # theoretical limit
```

A typical output is illustrated in Figure A.1. As expected, the relative frequency tends to $\frac{1}{2}$ as the number of trials increases.

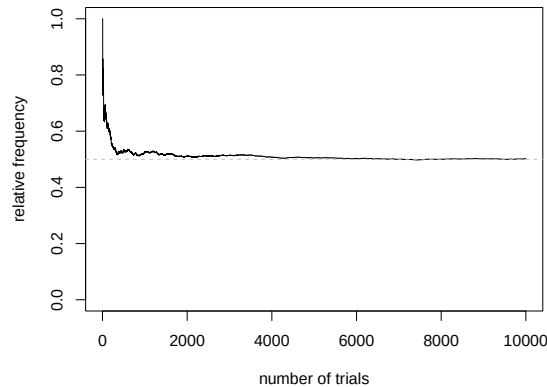


Figure A.1: A simulated trajectory of the relative frequencies of head when tossing a fair coin.

2. Following the frequentist interpretation, suppose we repeat the experiment n times under the same conditions. Since A implies B , we have

$$(\# \text{ of times } A \text{ occurs in } n \text{ trials}) \leq (\# \text{ of times } B \text{ occurs in } n \text{ trials}).$$

Dividing by n and letting $n \rightarrow \infty$, we have

$$\begin{aligned} \Pr(A) &= \lim_{n \rightarrow \infty} \frac{\# \text{ of times } A \text{ occurs in } n \text{ trials}}{n} \\ &\leq \lim_{n \rightarrow \infty} \frac{\# \text{ of times } B \text{ occurs in } n \text{ trials}}{n} \\ &= \Pr(B). \end{aligned}$$

3. In Table A.1 we compute the value of Y for each outcome. The possible values of Y are 0, 1 and 2. From the table, we have

$$\mathbb{P}(Y = 0) = \frac{2}{8}, \quad \mathbb{P}(Y = 1) = \frac{4}{8}, \quad \mathbb{P}(Y = 2) = \frac{2}{8},$$

and $\mathbb{P}(Y = y) = 0$ for all other values of y .

Prob	Outcome	Value of Y		
1/8	+	+	+	0
1/8	+	+	-	1
1/8	+	-	+	2
1/8	+	-	-	1
1/8	-	+	+	1
1/8	-	+	-	2
1/8	-	-	+	1
1/8	-	-	-	0
		2/8	4/8	2/8

Table A.1